UU Texture User Guide

Version 2.0 – February 2026

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UU Texture is a plugin that allows you to change an image or video on an object during PLAY, by simply giving a file path (local or network).

Goal: You type in a file address → press a button → the image/video appears.

1. Overview

The plugin offers two easy ways to work:

1) Blueprint nodes (to automate: only usable in play mode, also called runtime mode).
2) A ready-to-install Actor: UU Media Plate (to go fast: usable in editor mode).

Everything is designed to work in -game mode: no need to import media beforehand.

2. Installation

1) Open: Edit > plugins.
2) Active: UltiUnrealTexture.
3) Check that these Unreal plugins are active:

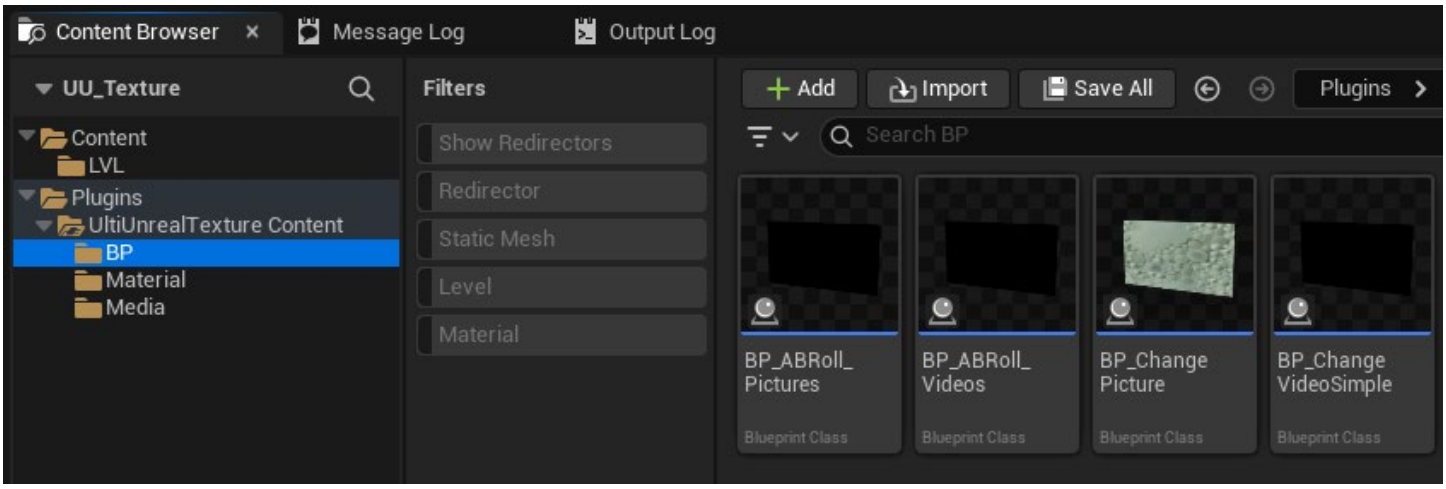
- Media Plate
- Media Framework Utilities (if available in your project)

4) Restart the project.

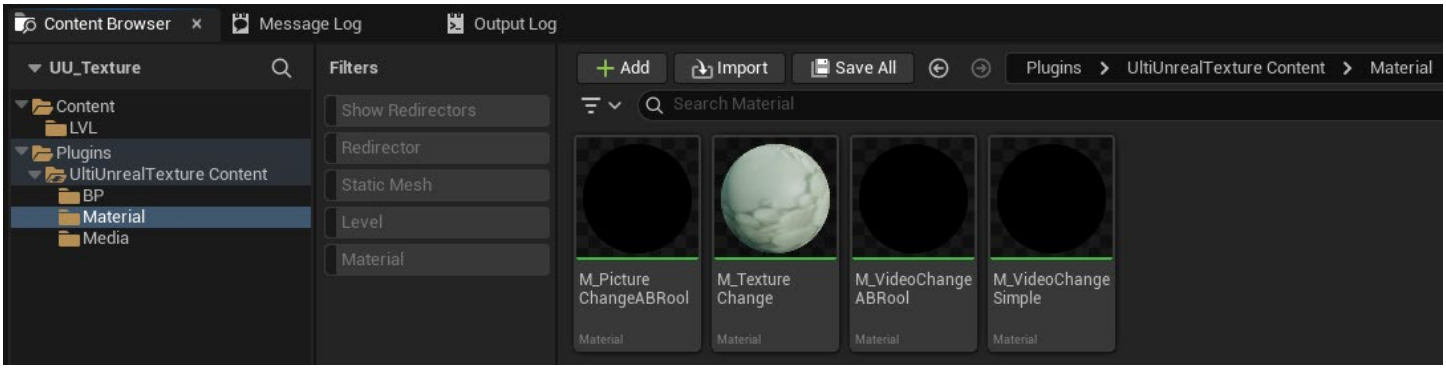
3. Plugin Content

In the Content Browser (Plugins) you will find:

- BP: Ready-to-use sample blueprints.
- Material: materials used by the examples.
- Media: Shared Media Players and Media Textures.



BP file – ready-to-ask examples.



Material folder – materials used by the examples.



Media folder – The Media Players and Media Textures used by the samples.

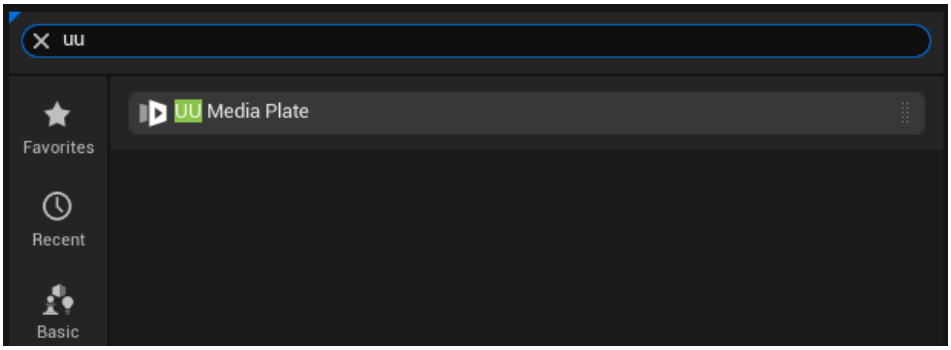
Important: The examples use the same Media Players / Media Textures. You can rename them, but if you change a name, the Blueprints/Materials must point to the new name.

4. Quick Use: UU Media Plate

UU Media Plate is the most user-friendly tool. It's a small TV: you give a video, then you press Cue / Play / Stop.

4.1 Placing it

1) In Place Actors, search for "uu".
2) Slide "UU Media Plate" into your Level.

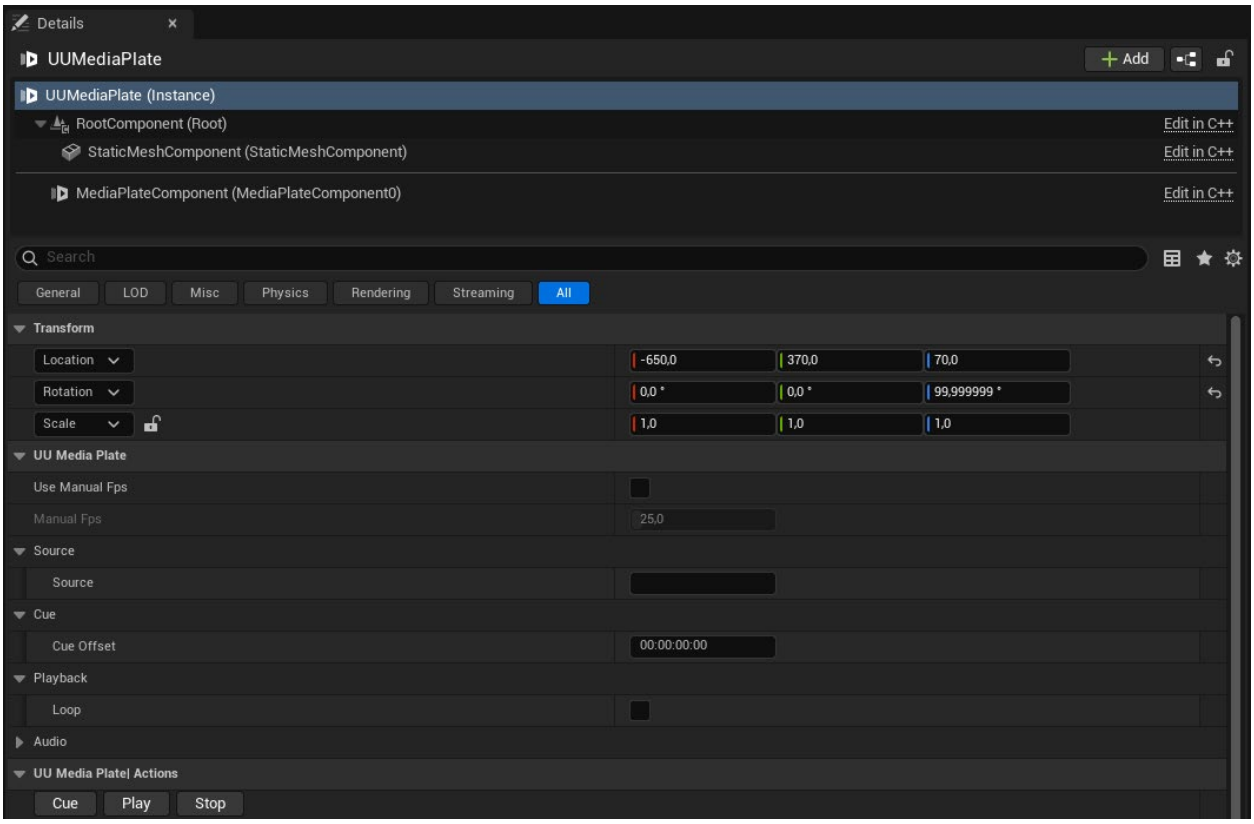


Place Actors: search for "uu" → UU Media Plate.

4.2 Fill it out

1) In Details, fill in the Source field (path of your video file).
2) Press Cue → then Play.
3) Press Stop to stop.

Tip: if you press play without Cue, it loads and then plays as soon as it's ready.



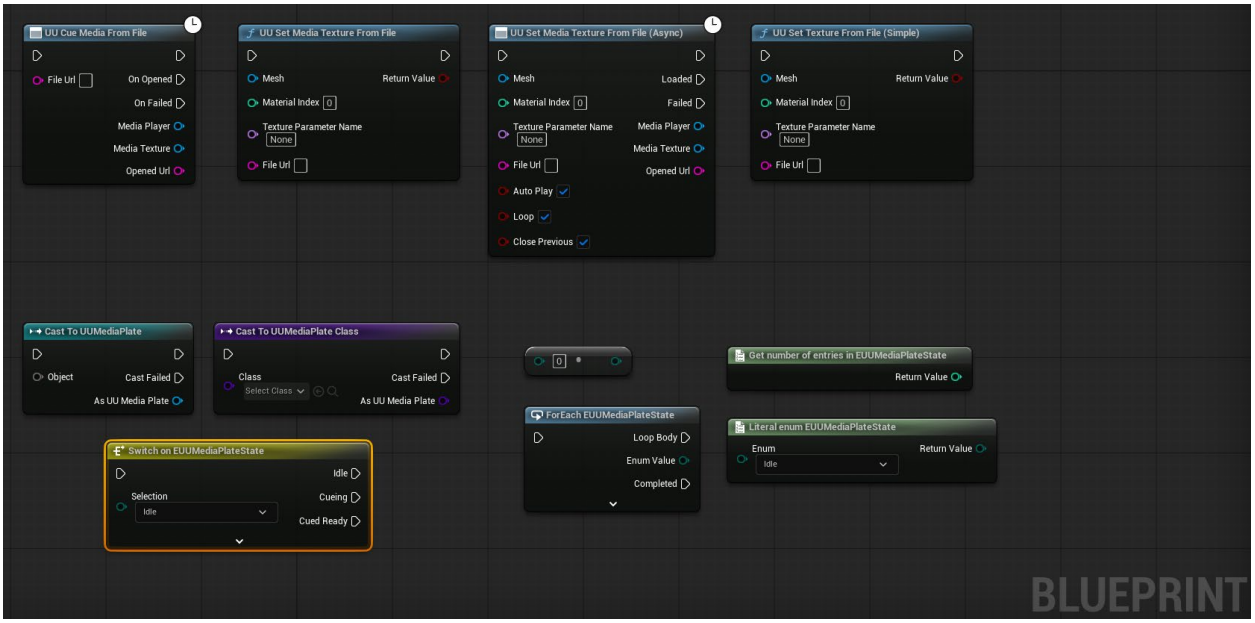
UU Media Plate – fields and buttons in Details.

4.3 Important fields (very simple explanation)

- Source: the address of the video.
  - Local example: D:/Videos/clip.mov
  - Network example: \\NAS\Videos\clip.mov
- Cue Offset: a simple offset before playing (handy if no timecode).
  - You write with periods, and the display is done with ":".
  - Example: 12 → 00:00:00:12 | 05.16 → 00:00:05:16
- Loop: Loops back.
- Audio: Keeps the sound (if the video has any).

5. Blueprint Nodes Available

Here, we explain "like a 2-year-old child": you plug in the node, you give an address, and it does the action.



UU Texture nodes visible in Blueprint.

5.1 UU Set Texture From File (Simple)

It's used to: change an IMAGE on an object (like pasting a photo).

You give:

- Mesh: the object (screen) where you want to put the image.
- Material Index: Which material of the object you want to change (often 0).
- Texture Parameter Name: the name of the "case texture" in your material.
- File Url: the address of your image file.

Return Value = true if all goes well.

5.2 UU Set Media Texture From File

It's used to: put a VIDEO on an object (like a TV).

You give the same thing (Mesh, Material Index, Texture Parameter Name) + File Url for the video.

5.3 UU Set Media Texture From File (Async)

It's for doing the same, but telling you when it's ready.

Outputs:

- Loaded: it's ready.
- Failed: it didn't work.

5.4 UU Cue Media From File

It's used to: prepare the next video BEFORE playing it.

It's like: "I'm charging now, so Play goes faster".

Outputs:

- On Opened: ready.
- On Failed: not ready.

5.5 Summary of Nodes

The "Cast To ..."

Cast To UUMediaPlate

It's like saying:
"The thing I have in my hand... is it really a UU Media Plate?"

- If so→ you can use its functions/variables.
- If not→ it's fine in Cast Failed (so you know it's not the right guy).

Cast To UUMediaPlate Class

Same idea, but instead of an object already placed, you start with a class (a "model").
"This class... does it correspond to UU Media Plate?"

The "Switch on..." node

Switch on EUUMediaPlateState

It's a train switch.
You have a "State" variable, and depending on its value you go to a different output:

- Idle: nothing happens (at rest)
- Cueing: Cue
- Cued Ready: Ready to Play (Cue Completed)

Very useful for: displaying a message, preventing a double click, or changing the UI according to the state.

The "Enum" nodes

Literal enum EUUMediaPlateState

It's just a "fixed label". "I explicitly mean: Idle" or "I want to say explicitly: Cueing". It's useful when you want to compare or force a value.

Get number of entries in **EUUMediaPlateState.** It says: "How many possible values are there in this state?"
Example: Idle / Cueing / Cued Ready → that would be 3. It's especially useful if you're making menus, lists, or loops.

ForEach EUUMediaPlateState

It's: "I'm taking every possible value of EUUMediaPlateState, one by one." So it's going to do: Idle → then Cueing → then Cued Ready → etc. Very useful for automatically generating a list, a debug, or a drop-down menu.

The **small node "0"** in the middle (constant)
The node with 0 is just: "Here is the number zero." Often this is used for a Material Index = 0, or a default value.

What is the purpose of these "non-UU" nodes in practice?

- Cast: to make sure we're talking about the right actor (UU Media Plate).
- Switch on State: to react differently depending on whether it's ready, charging, etc.
- Enum / ForEach: to properly manipulate possible states.
- 0: base value when you want to point to the "first" element (often the first material).

The nodes that start with "UU ..." = these are the magic buttons:
"Load video", "Change image", etc.

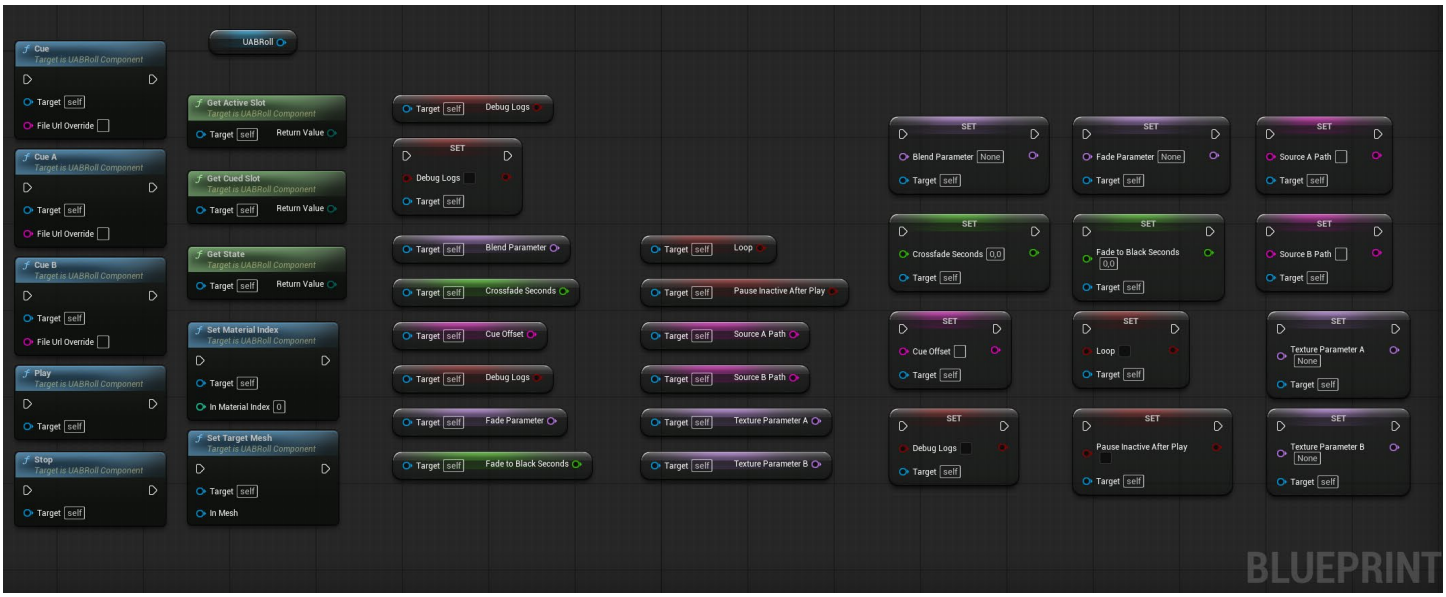
Nodes like Cast, Switch, Enum = they are road signs:
"Is this the right object?", "What state are we in?", "What to do now?"
Basically: UU does, the others organize.

6. AB Roll Video

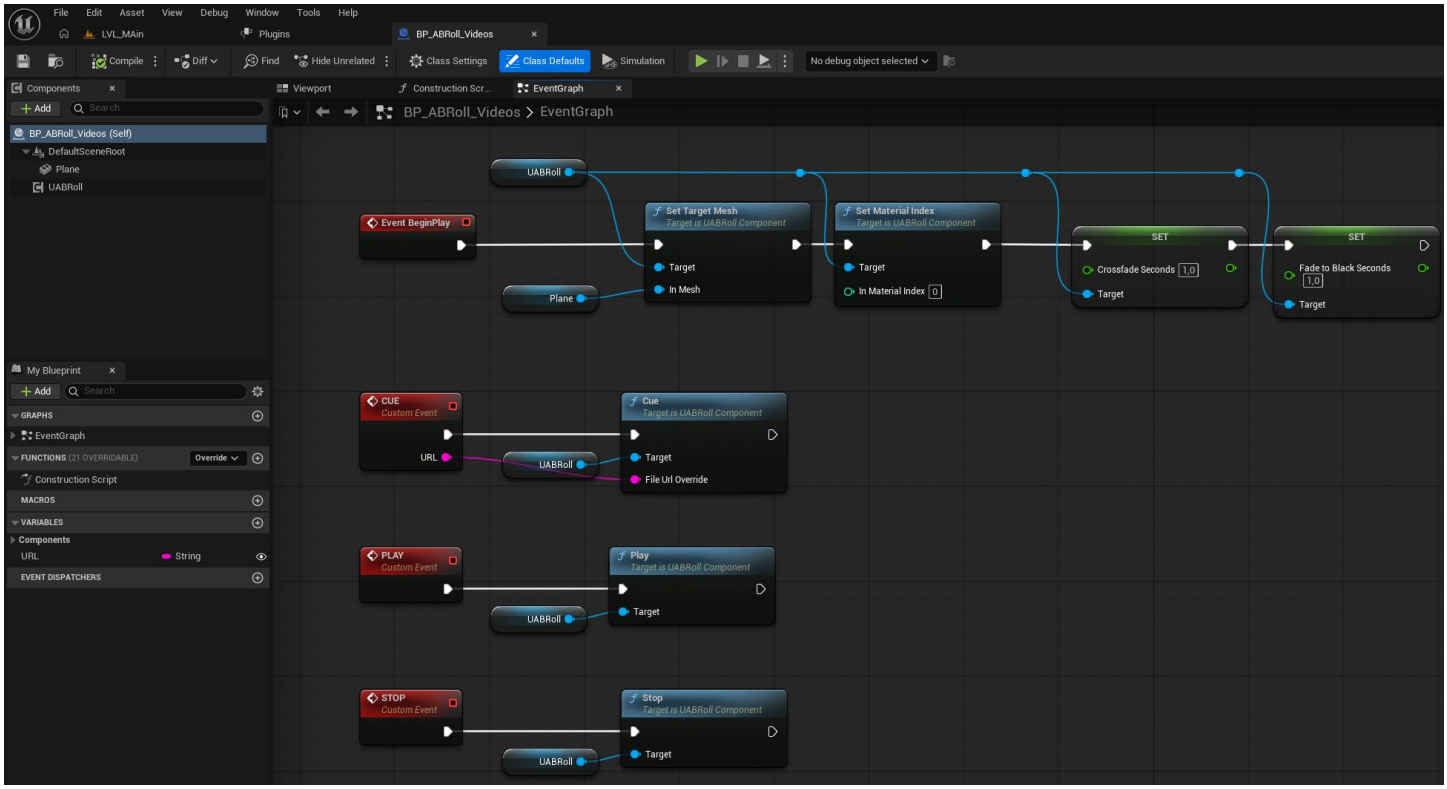
A sample Blueprint named BP_ABRoll_Videos is included with the plugin.

6.1 BP_ABRollScreen

It's used to: make a clean A/B Roll (Cue → Play → Stop).



1- Add a mesh in your blueprint and **assign it the material M_VideoChangeABRool included with the plugin**
2- Add the UABRoll component
3- Reproduce the blueprint below (included in the plugin)



Principle: "real AB Roll"

The UABRollComponent implements a two-source AB Roll (A and B) with a "black" step:
• STOP → fade to black (and stay black).
• CUE → prepares the next source (A or B) without changing the live image.
• PLAY→ fade between the visible source and the cue source (A ↔ B).
• If you are already black (after STOP), PLAY = fade black → video (without showing an A↔B before).

Prerequisites (materials)

The component drives a Material Instance Dynamic (MID) over a mesh. The material must expose:• 2 Texture parameters (slot A and slot B)• 2 Scalar parameters: one for the A/B mix (0→A, 1→B) and one for the black fade (0→image, 1→black)

If you use the master material from the video demo (with MediaTextureA/MediaTextureB, BlendAB, FadeToBlack), you can keep the default settings.

Set Target Mesh



Assigns the mesh that the component should drive (creation/refresh of the MID on this mesh).
Starters:
• In Mesh (MeshComponent): e.g. your Plane / StaticMeshComponent.
Outputs:
• None.
Simple example:
• In BeginPlay: Call Set Target Mesh with your Plane reference.
• Only then: Cue / Play / Stop.
Notes:
• The material must be present on the mesh's MaterialIndex slot.
• If you change the material at runtime (or via Remote Control), the component can recreate the MID.

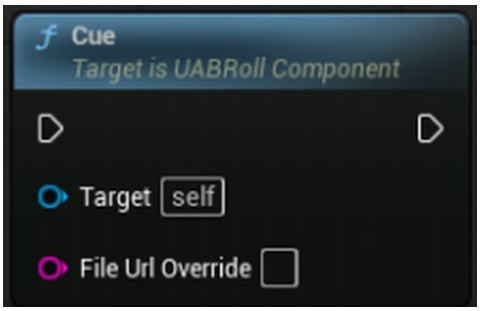
Set Material Index



Chooses the slot index on which the component should create/control the MID.
Starters:

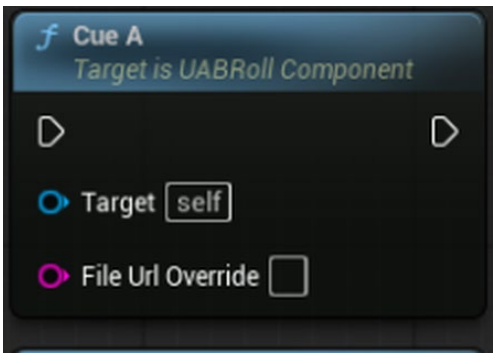
• In Material Index (int): 0 by default in most cases.
Outputs:
• None.
Simple example:
• Plane → Set Material Index = 0 (if your material is on slot 0).
Notes:
• If the index does not match, you will see "nothing changes". Enable Debug Logs for diagnostics.

Cue



Prepares the next source without changing the live image. The component automatically chooses the inactive slot (A or B).
Starters:
• File Url Override (string, optional): The path of the video file to be uploaded for this CUE. If empty, the component can use Source A Path / Source B Path depending on the targeted slot.
Outputs:
• None (but you can listen to the success/error events if exposed in your BP).
Simple example:
• From UltiUnreal/WRC: create a Custom Event 'CUE(URL)' and link it to this node.
• In-game: call Cue("C:/Videos/shot01.mp4") then Play.
Notes:
• CUE should not cut the image: it loads/seeks the source and pauses it until you do PLAY.
• Best practice: use Cue(File Url Override) rather than modifying hot path variables.

Cue A



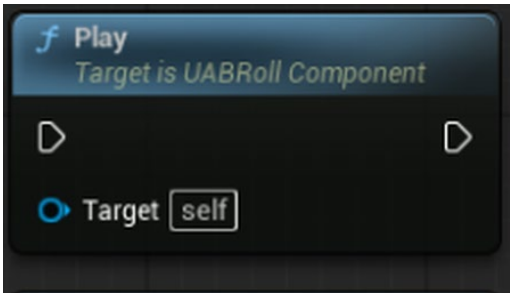
Force the CUE into slot A, regardless of which slot is currently visible.
Starters:
• File Url Override (string, optional).
Outputs:
• None.
Simple example:
• At startup: Cue A("C:/Videos/A.mp4") then Cue B("C:/Videos/B.mp4") to preload the 2.
Notes:
• Useful especially for testing/setup. In production, you use Cue (auto).

Cue B



Force the CUE into slot B, regardless of which slot is currently visible.
Starters:
• File Url Override (string, optional).
Outputs:
• None.
Simple example:
• Preload B without impacting what's live: Cue B("C:/Videos/alt.mp4").

Play



Make the transition:• If you are in black → fade → cue source.• Otherwise → crossfade A↔B between the visible source and the cue source.
Starters:
• None.
Outputs:
• None.
Simple example:
• Cue(URL) → Play (makes the transition).
• After Stop (black): Cue(URL) → Play (black → video).
Notes:
• Time depends on Crossfade Seconds (A↔, B) and Fade to Black Seconds (video black ↔).

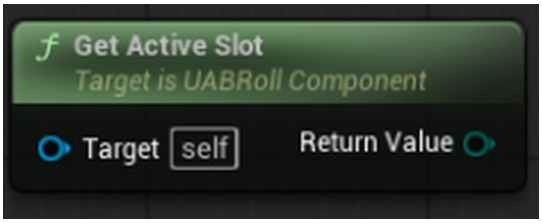
Stop



Fade to black and stay black (until the next Play).
Starters:

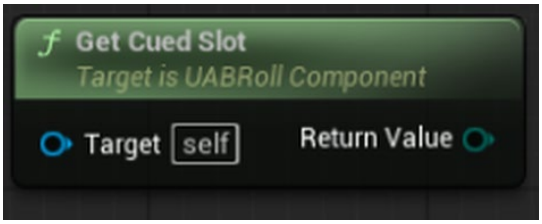
• None.
Outputs:
• None.
Simple example:
• Stop during live → black screen.
• Then Cue(URL) → Play to go back to a source.

Get Active Slot



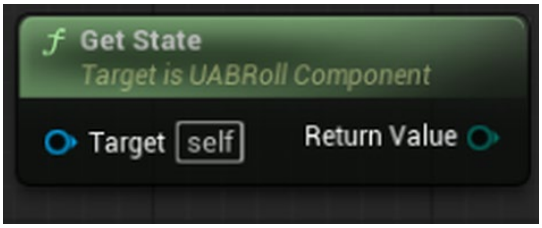
Flips the currently visible slot (A or B).
Outputs:
• Return Value (Enum EUUABRollSlot).
Simple example:
• Display in a Print String for debug: ActiveSlot.

Get Cued Slot



Returns the slot currently prepared by CUE (the one that PLAY will take).
Outputs:
• Return Value (Enum EUUABRollSlot).
Simple example:
• Debug: check that the cue slot is inactive before playing.

Get State

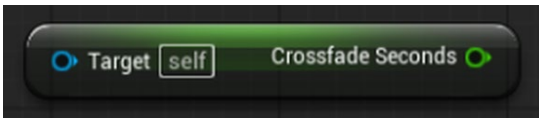


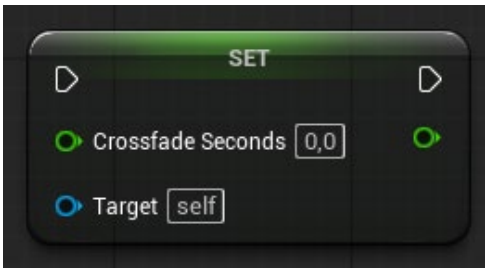
Returns the internal state of the component (idle, cueing, transition, black, error...).
Outputs:
• Return Value (Enum EUUABRollState).
Simple example:
• If State == Cueing: avoid spam Play, wait for the loading to finish.
• If State == Error: Display an operator message.

Settings (Get / Set)

The following parameters exist in both Get and Set. They are useful in BeginPlay (configuration) or via Remote Control.

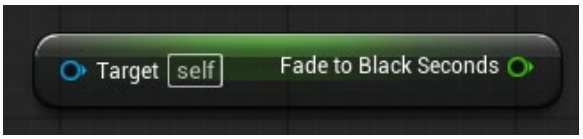
Crossfade Seconds

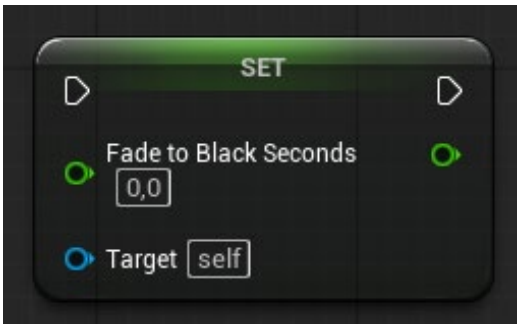




The duration (in seconds) of the fade A↔B when PLAY (when not in black). Typical value: 0.5 to 2.0 s
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• In BeginPlay: Set Crossfade Seconds = 1.0.

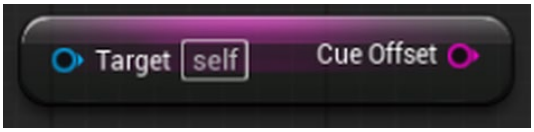
Fade to Black Seconds





The duration (in seconds) of fades to/from black (STOP and PLAY from black). Typical value: 0.3 to 1.0 s
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• In BeginPlay: Set Fade to Black Seconds = 0.5.

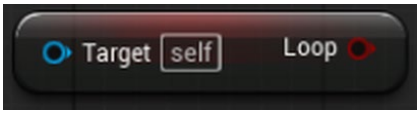
Cue Offset





The read lag (in seconds) applied at the time of the CUE (initially seek). Typical value: 0.0 s
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• To skip 2 seconds at the beginning: Set Cue Offset = 2.0.
Notes:
• Useful if your media has a black first frame or a clapperboard to ignore.

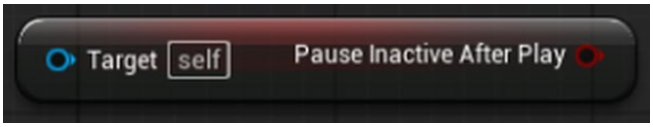
Loop

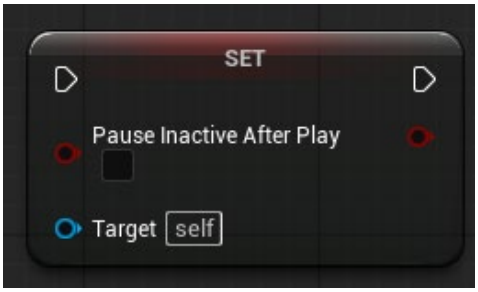




Enables/disables video looping. Typical value: false (often), or true for wrapping loops
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• Set Loop = true for background video.

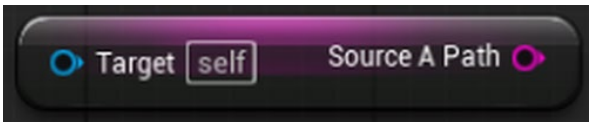
Inactive Break After Play

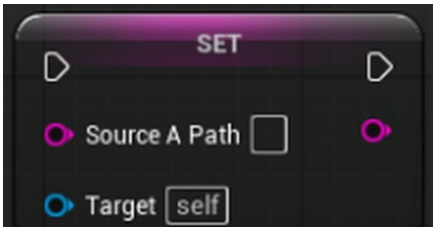




If enabled, the slot that becomes inactive after PLAY is paused (CPU/GPU saving). Typical value: true
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• Set Pause Inactive After Play = true (recommended).
Notes:
• If you want videos that keep running in the background (rare), turn it off.

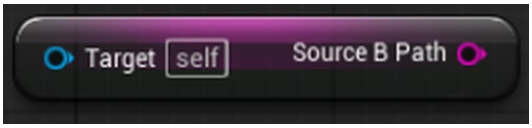
Source A Path

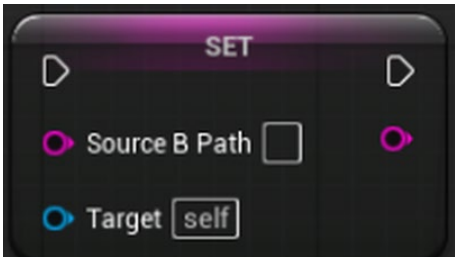




The default path of the video file in slot A (used if Cue/CueA does not have an override). Typical value: Recommended absolute path
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• Set Source A Path = C:/Videos/A.mp4.
Notes:
• In live control, you can skip these fields and skip the path directly into Cue(File Url Override).

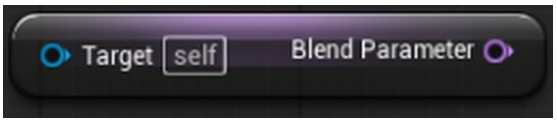
Source B Path

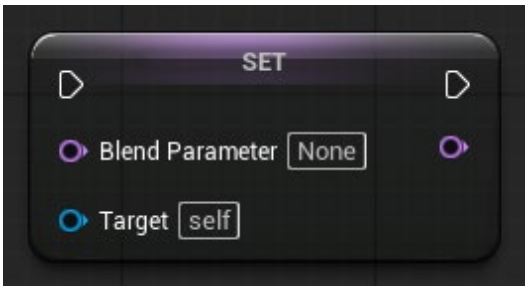




Default path of the video file in slot B (used if Cue/CueB does not have an override). Typical value: Recommended absolute path
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• Set Source B Path = C:/Videos/B.mp4.

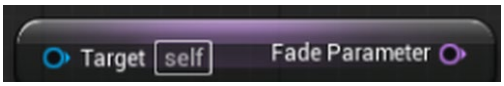
Blend Parameter

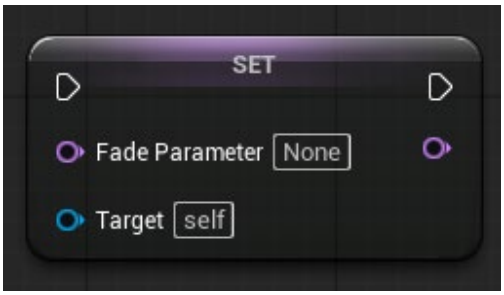




The name of the Scalar parameter in the material that controls the A/B mix (0→A, 1→B). Typical Value: BlendAB (demo)
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• If your material uses 'uu_blend': Set Blend Parameter = uu_blend.
Notes:
• If this name does not match, PLAY will not transition (the material does not react).

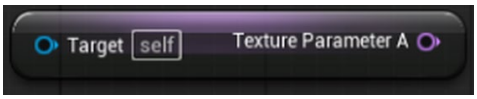
Fade Parameter

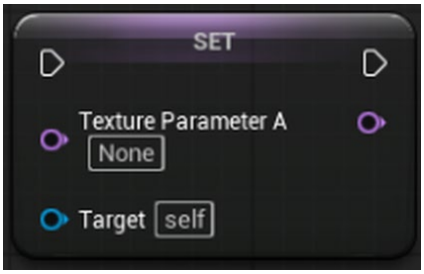




The name of the Scalar parameter in the material that controls fade to black (0→frame, 1→black). Typical Value: FadeToBlack (Demo)
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• If your material uses 'uu_fade': Set Fade Parameter = uu_fade.

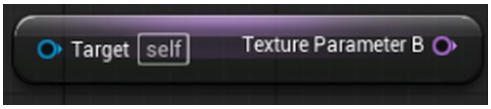
Texture Parameter A

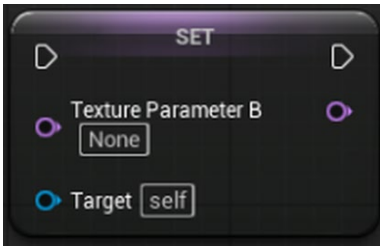




The name of the Texture in Material parameter for the Slot A texture (MediaTexture A). Typical Value: MediaTextureA (Demo)
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• If your material uses 'Tex_A': Set Texture Parameter A = Tex_A.

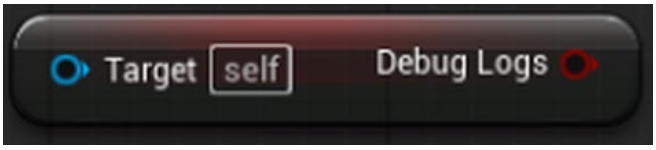
Texture Parameter B

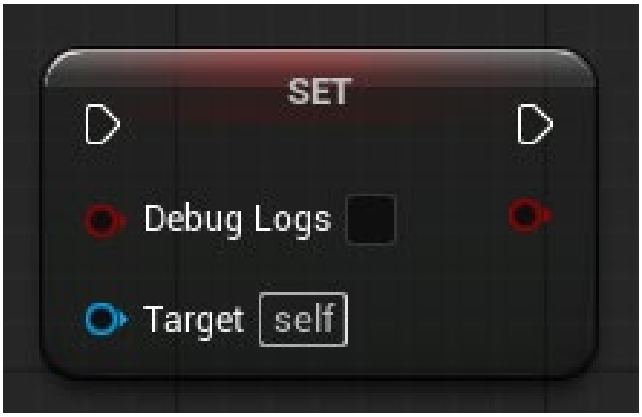




The name of the Texture in Material parameter for the Slot B Texture (MediaTexture B). Typical value: MediaTextureB (demo)
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• If your material uses 'Tex_B': Set Texture Parameter B = Tex_B.

Debug Logs





Activates logs (UE_LOG) to diagnose: missing MID, missing material parameters, files not found, etc. Typical value: false in prod, true in debug
Starters:
• (Set)—The value to assign.
Outputs:
• (Get)—The current value.
Simple example:
• In case of a problem: Set Debug Logs = true then look at Output Log.

Full example (minimum BP)

The objective: to control a Plane and make a transition between two videos via Remote Control or BP.
1) In the BP Actor: add a StaticMeshComponent (Plane) and add a UABRollComponent.
2) On the Plane: assign the AB Roll master material (the same as for the video).
3) In Event BeginPlay: Set Target Mesh (Plane) → Set Material Index (0) → Set Crossfade Seconds (1.0) → Set Fade to Black Seconds (0.5).
4) To control: CUE(URL) → PLAY(transition); STOP (black); STOP → CUE(URL) → PLAY (black → video).

UltiUnreal/WRC Tip: Exposes Custom Events 'CUE(URL)', 'PLAY', 'STOP' and calls the component nodes inside. Avoids directly modifying hot path variables: always goes through Cue(File Url Override).

6.2 BP_BP_ABRoll_Pictures

The operation is exactly the same as for a Video AB Roll, **use the UABPhotoRoll component and apply the M_PictureChangeABRool material to your Mesh.**

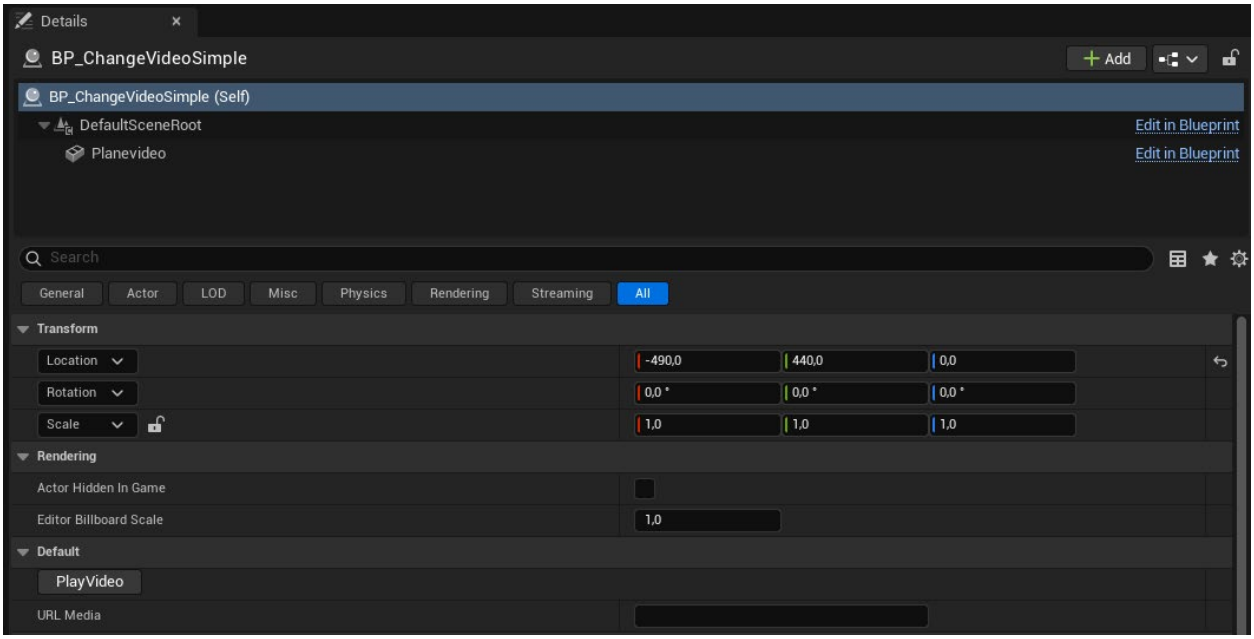
6.3 BP_ChangeVideoSimple

It's used to: play a video with ONE button.

In Details:

• Media URL: The address of the video.

Button: PlayVideo (load + play).



BP_ChangeVideoSimple – a button to launch.

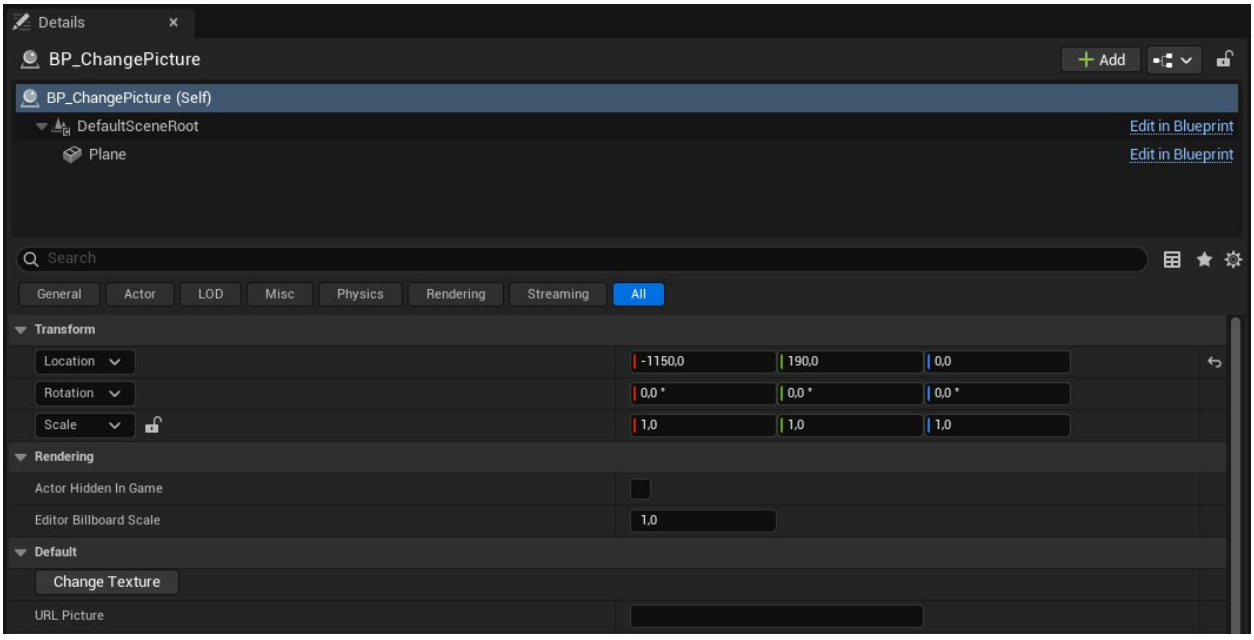
6.4 BP_ChangePicture

It's used to: change an image with ONE button.

In Details:

- URL Picture: The address of the image.

Button: Change Texture.



BP_ChangePicture – changes an image from a path.

7. Formats & Tips

Formats used in your workflow: mov, mpg, mpeg, HAP (depending on the available codecs).

MKV: sometimes yes, sometimes no (it depends on the backend and codecs). The best test: try it out in your project.

Network: make sure that the machine has the rights to share it (otherwise the file is "not found").

8. Support & Updates

UU Texture is part of the UltiUnreal toolset developed by JDarrasVP for multi-station virtual production workflows. Future updates may add new options, presets, or integration aids for specific broadcast environments.

Feel free to ask your questions on my YouTube page at http://youtube.com/jdarrasvp or on my website: www.jdarrasvp.fr